

Statistics & Data Analysis

Comprehensive Midterm Cheat Sheet · Probability → Random Variables → Distributions → Normal & CLT

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Classes 01–04

How to use: Formulas sit in blue boxes; **METHOD** notes in green; **TRAP** = the mistakes that cost points. Every rule is cross-checked against the sample-midterm & HW solution keys. The fully-worked Sample Midterm is §9.

Probability

Conditional, Bayes, unions, total prob., independence, tables

Random Variables

PMF/CDF, $E(X)$, Var, σ ; $E(XY)$; rules for $aX+bY$

Distributions

Binomial, Poisson, Hypergeometric, Covariance/Correlation

Normal & CLT

z-scores, z-table, sum vs. mean, normal=binomial

§1 Probability Foundations

Vocabulary

- Experiment** → process with chance outcomes. **Sample space (S)** = set of *all* outcomes. **Event** = a subset of outcomes (capital letters A, B, ...).
- Simple events** = the most basic, mutually-exclusive outcomes; S is the collection of all of them. *Ex:* two coins → $S = \{HH, HT, TH, TT\}$.
- Mutually exclusive** = both cannot happen at once ($A \cap B = \emptyset$).
- Intersection $A \cap B$** = both occur. **Union $A \cup B$** = A or B or both.

The Core Probability Rules

RANGE & TOTAL

$$0 \leq P(A) \leq 1 \cdot P(S) = 1 \cdot \sum p(\text{all outcomes}) = 1$$

COMPLEMENT

$$P(A) = 1 - P(\bar{A})$$

EQUALLY-LIKELY OUTCOMES

$$P(A) = m / M \quad (m \text{ favorable of } M \text{ equally-likely})$$

ADDITION (GENERAL)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

ADDITION (MUTUALLY EXCL.)

$$P(A \cup B) = P(A) + P(B)$$

MULTIPLICATION

$$P(A \cap B) = P(A|B) \cdot P(B) = P(B|A) \cdot P(A)$$

INDEPENDENT EVENTS

$$P(A \cap B) = P(A) \cdot P(B)$$

Conditional Probability & Independence

CONDITIONAL

$$P(A|B) = P(A \cap B) / P(B)$$

Independence test — A, B independent iff any one holds:

$$P(A|B) = P(A) \Leftrightarrow P(B|A) = P(B) \Leftrightarrow P(A \cap B) = P(A)P(B)$$

METHOD

To decide independence: compute $P(A) \cdot P(B)$ and compare to the given $P(A \cap B)$. Equal → independent; not equal → dependent.

TRAP

Mutually exclusive $\neq$ independent. If A, B are mutually exclusive with $P>0$ they are *dependent* (knowing A happened forces $P(B)=0$). Conditioning *shrinks* the sample space.

Total Probability & Bayes' Theorem

LAW OF TOTAL PROBABILITY

$$P(A) = P(A|B) \cdot P(B) + P(A|\bar{B}) \cdot P(\bar{B})$$

BAYES' THEOREM

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A|B) \cdot P(B) + P(A|\bar{B}) \cdot P(\bar{B})}$$

METHOD

Two-stage / "jar" & testing problems: Stage 1 = which path (die roll, coin, jar). Stage 2 = outcome given the path. Multiply along a branch; add branches for total probability; Bayes to reverse (find the cause given the effect). A **tree** or a **frequency table scaled to 10,000** both work.

EX · DISEASE TEST (BAYES)

$P(\text{Dis}) = .01$, $P(\text{Pos}|\text{Dis}) = .98$, $P(\text{Neg}|\bar{\text{Dis}}) = .99 \Rightarrow P(\text{Pos}|\bar{\text{Dis}}) = .01$. Find $P(\text{Dis}|\text{Pos})$.

$$P(\text{Dis}|\text{Pos}) = (.98)(.01) / [(.98)(.01) + (.01)(.99)] = .0098 / .0197 = \approx .498$$

Table of 10,000: 98 true-pos, 99 false-pos → $98/197 = .498$.

Contingency (Two-Way) Tables

Joint = one cell ÷ grand total. **Marginal** = row/col total ÷ grand total. **Conditional** = cell ÷ that row-or-column total (the "given" resets the denominator).

	USA	Non-US	Total
Male	22	18	40
Female	13	9	22
Total	35	27	62

$P(\text{Male}|\text{Non-US}) = (40+27-18)/62 = 49/62$. $P(\text{Asian}|\text{Female}) = \text{cell}/\text{row} = 3/22$. $P(\text{Male}|\text{not-Euro}) = 35/54$.

§2 Counting: Factorials, Permutations, Combinations

FACTORIAL

$$x! = x(x-1) \cdots (2)(1) \cdot 0! = 1$$

PERMUTATIONS — ORDER MATTERS

$${}_n P_k = n! / (n-k)!$$

COMBINATIONS — ORDER DOESN'T

$$C(n,k) = {}^n C_k = n! / [k!(n-k)!]$$

- Order x distinct items in a row → $x!$

- Pres/VP/Treas from 10 (order matters) → $10 \cdot 9 \cdot 8 = 720 = 10!/7!$
- Committee of 3 from 8 (no order) → $C(8,3) = 56$
- Choose 2 of 5 firms, $P(\text{they're A\&B}) = 1/C(5,2) = 1/10$; $P(\text{E among 3 chosen of 5}) = C(4,2)/C(5,3) = 6/10 = 3/5$

METHOD

"Arrange / rank / assign titles" → permutation. "Choose / select / committee / sample" → combination.

§3 Random Variables: PMF, E(X), Variance

Definitions

- Random variable (X)** = takes numeric values by chance. **Discrete** = countable ("counting"); **Continuous** = measured, in-between values.
- PMF / pdf:** $p(x)=P(X=x)$. Requires $0 \leq p(x) \leq 1$ and $\sum p(x) = 1$.
- CDF:** $F(x)=P(X \leq x)$. Then $P(X > x) = 1 - F(x)$; $P(a \leq X \leq b) = F(b) - F(a-1)$ for integers. Get pmf from cdf by differencing.

Expected Value & Variance

MEAN / EXPECTED VALUE

$$\mu_X = E(X) = \sum x \cdot p(x)$$

VARIANCE — DEFINITION

$$\sigma_X^2 = \text{Var}(X) = \sum (x - \mu)^2 \cdot p(x)$$

VARIANCE — SHORTCUT (USE THIS!)

$$\text{Var}(X) = \sum x^2 \cdot p(x) - \mu^2 = E(X^2) - [E(X)]^2$$

STANDARD DEVIATION

$$\sigma_X = \sqrt{\text{Var}(X)} \quad (\text{always } \geq 0; \text{ in original units})$$

METHOD Build a table: columns x , $p(x)$, $x \cdot p(x)$, $x^2 \cdot p(x)$. Sum col 3 $\rightarrow E(X)$. Sum col 4 $\rightarrow E(X^2)$. $\text{Var} = E(X^2) - [E(X)]^2$.

Rules for E and Var (a, b constants)

$$E(a) = a$$

$$\text{Var}(a) = 0$$

$$E(aX+b) = a \cdot E(X) + b$$

$$\text{Var}(aX+b) = a^2 \cdot \text{Var}(X)$$

$$E(X+Y) = E(X)+E(Y) \text{ always}$$

$$\sigma_{aX} = |a| \cdot \sigma_X$$

TRAP Adding a constant **b** shifts the mean but **NOT** the variance ($\text{Var}(X+b)=\text{Var}(X)$). A multiplier **a** comes out squared in variance (a^2) but plain in the mean.

METHOD · EMPIRICAL RULE FOR RVS $\approx 2/3$ of values within $\mu \pm \sigma$; $\approx 95\%$ within $\mu \pm 2\sigma$.

§4 Covariance, Correlation & Sums of Random Variables

COVARIANCE

$$\text{Cov}(X,Y) = E(XY) - \mu_X \mu_Y = E[(X - \mu_X)(Y - \mu_Y)]$$

CORRELATION ($-1 \leq \rho \leq 1$)

$$\rho_{XY} = \text{Cov}(X,Y) / (\sigma_X \cdot \sigma_Y) \Rightarrow \text{Cov}(X,Y) = \rho_{XY} \cdot \sigma_X \cdot \sigma_Y$$

$E(XY)$

$$E(XY) = \sum (x \cdot y) \cdot P(x,y) \quad \text{sum over joint outcomes}$$

Variance of a Sum / Combination

TWO VARIABLES

$$\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y) + 2 \cdot \text{Cov}(X,Y)$$

DIFFERENCE

$$\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y) - 2 \cdot \text{Cov}(X,Y)$$

GENERAL WEIGHTED

$$\text{Var}(aX+bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \cdot \text{Cov}(X,Y)$$

SCALING COVARIANCE

$$\text{Cov}(aX,bY) = ab \cdot \text{Cov}(X,Y) \quad \text{Cov}(X,X) = \text{Var}(X)$$

INDEPENDENT $\Rightarrow$ COV = 0 Then $\text{Var}(X+Y) = \text{Var}(X) + \text{Var}(Y)$ and $\text{Var}(X-Y) = \text{Var}(X) + \text{Var}(Y)$ too. **Variances add; standard deviations do NOT** ($\sigma_{X+Y} \neq \sigma_X + \sigma_Y$).

EX · COVARIANCE DRILL

$$\sigma_X=10, \sigma_Y=12, \sigma_Z=20; \rho_{XY}=.8, \rho_{XZ}=-.5.$$

$$\text{Cov}(X,Y) = .8 \cdot 10 \cdot 12 = \mathbf{96}; \text{Cov}(X,Z) = -.5 \cdot 10 \cdot 20 = \mathbf{-100}; \text{Cov}(2X,3Y) = 6 \cdot 96 = \mathbf{576}.$$

$$\text{Var}(X+Y) = 100 + 144 + 2(96) = \mathbf{436}. \text{Var}(3X+5Z) = 9(100) + 25(400) + 2 \cdot 15(-100) = \mathbf{7900}.$$

EX · PORTFOLIO (WEIGHTS SUM TO 1)

$$\text{Return} = w_1 X + w_2 Y. w = .4/.6, E = .12/.19, \sigma = .09/.28, \rho = -.4.$$

$$E(R) = .4(.12) + .6(.19) = \mathbf{.162}$$

$$\text{Var} = .4^2(.09^2) + .6^2(.28^2) + 2(.4)(.6)(-.4)(.09)(.28) = .0247 \rightarrow \sigma = \mathbf{.157}$$

Negative ρ lowers portfolio risk — the diversification effect.

§5 The Named Discrete Distributions

Binomial (n, π) — fixed # of independent trials

Use when: n fixed trials · each success/fail · constant π · independent (**sampling WITH replacement**). $X = \#$ successes $= Y_1 + \dots + Y_n$, a sum of n **Bernoulli** trials (one trial: $E = \pi$, $\text{Var} = \pi(1-\pi)$).

$$P(X=k) = C(n,k) \cdot \pi^k \cdot (1-\pi)^{n-k}$$

$$E(X) = n\pi \quad \text{Var}(X) = n\pi(1-\pi) \quad \sigma = \sqrt{n\pi(1-\pi)}$$

"At least one" $= 1 - P(0) = 1 - (1-\pi)^n$. Excel: =BINOM.DIST(k, n, π , FALSE) for $P(X=k)$; TRUE for $P(X \leq k)$.

EX · SALESPERSON, N=25 VISITS, $\pi=.3$

$$P(8 \text{ sales}) = C(25,8) \cdot (.3)^8 \cdot (.7)^{17} = \mathbf{.165}. E = 25 \cdot (.3) = \mathbf{7.5}; \sigma = \sqrt{25 \cdot .3 \cdot .7} = \mathbf{2.29}.$$

Poisson (λ) — random events over time/space

Use when: counting events that occur one-at-a-time, independently, at a steady rate; no upper limit on x.

$P(X=k) = e^{-\lambda} \cdot \lambda^k / k! \quad (k=0,1,2,\dots)$

$E(X) = \text{Var}(X) = \lambda \cdot \lambda = \text{rate} \times \text{duration}$

METHOD Scale λ to the window asked: 4/hr over 2 hr → λ=8; over 15 min → λ=1. "At least one" = 1 - e^{-λ}.

EX · CALLS AT 4/HR (POISSON)
P(5 in 2 hr): λ=8 → e⁻⁸·8⁵/5!=**.0916** . P(≥1 in 1 hr): λ=4 → 1-e⁻⁴=**.982** .

Hypergeometric (N, r, n) — sampling WITHOUT replacement

$P(X=x) = \frac{C(r,x) \cdot C(N-r, n-x)}{C(N,n)}$

N=population, r=successes in pop, n=sample, x=successes in sample. **Without replacement** (draws not independent) → hypergeometric, not binomial.

§6 The Normal Distribution & z-Scores

Properties

- Continuous, **bell-shaped, symmetric** about μ. Two parameters: mean μ, variance σ².
- Only the **number of SDs from the mean** matters — that's what standardizing captures.
- Continuous ⇒ P(X=a)=0, so **< and ≤ are interchangeable**.

STANDARDIZE
 $Z = (X - \mu) / \sigma \rightarrow Z \sim \text{Normal}(0, 1)$

INTERVAL
 $P(a \leq X \leq b) = P(X \leq b) - P(X \leq a)$

"FIND THE VALUE" / PERCENTILE
 $x = \mu + z \cdot \sigma$ (z from the target tail area)

Continuous RV basics

Probability = **area under the density curve**; any single point has probability 0. Notation **X ~ N(μ, σ²)** (2nd parameter is the *variance*). **Uniform** on [1,5]: height = 1/(5-1) = ¼ → P(2≤X≤4)=2·¼=½.

EX · 20 MARBLES (12 RED), DRAW 5, NO REPLACEMENT

P(exactly 3 red)=C(12,3)·C(8,2)/C(20,5)=**.397** (with replacement → Binomial: C(5,3)·6³·4²=.346).

Which distribution? — quick selector

Situation	Model
Fixed n trials, constant p, w/ replacement	Binomial
Draw without replacement from finite pop	Hypergeom.
Events per time/area, steady rate	Poisson
Sum/mean of many RVs; large n	Normal (CLT)

APPROX Hypergeometric ≈ Binomial when n is small vs. N (pop large):
P(none)≈(1-r/N)ⁿ.

Reading the class z-table

This course's table gives **A(z)=P(0≤Z≤z)** (area from the middle out). Convert:

- P(Z≤z) = **0.5 + A(z)** (z>0)
- P(Z>z) = **0.5 - A(z)**
- P(Z≤-z) = 0.5 - A(z) (**symmetry**)
- P(-a≤Z≤b) = A(a)+A(b)

METHOD Always draw the bell, mark μ, shade what's asked, then translate to A(z) pieces. "Find k with P(0≤Z≤k)=.40" → read the table backwards → k≈1.28.

Some tables (& Excel NORM.S.DIST) give the full cumulative Φ(z)=P(Z≤z) directly — then read it straight off, no +.5.

INTERPOLATE Between rows: z ≈ z_{lo} + [(target-A_{lo})/(A_{hi}-A_{lo})]·(z_{hi}-z_{lo}). Ex: Φ=.90 lies between .8997 (1.28) & .9015 (1.29) → k≈1.282.

EX · X ~ N(100, 10²)
P(92<X<104)=P(-.8<Z<.4)=Φ(.4)-Φ(-.8)=.6554-.2119=**.4435**

EX · COKE DEMAND ~ N(150, 20)
P(X≤200): z=(200-150)/20=2.5 → .5+.4938=**.9938**
P(X≥180): z=1.5 → .5-.4332=**.0668**
Stock so P(run out)=.1: z=1.28 → 150+1.28(20)=175.6 → stock **176** .

§7 Sampling Distributions & the Central Limit Theorem

CLT: the **sum** (or **average**) of many independent, identically-distributed RVs is ≈ **Normal** — even when the underlying variable is not normal. Conditions: iid + independent + large n (**rule of thumb n ≥ 30**).

SUM OF N	AVERAGE OF N ($\bar{X}$)
mean = nμ	mean = μ
SD = σ√n	SE = σ/√n
Z = (T - nμ)/(σ/√n)	Z = ($\bar{X}$ - μ)/(σ/√n)

TRAP **Sum vs. average.** Sum spreads out (σ√n); the average tightens (σ/√n). Pick the right one and don't mix variances with SDs.

Two special cases

- Underlying is Normal** → the sum/average is **exactly** Normal for any n (not an approximation).
- Normal = Binomial:** valid when **nπ > 5 and n(1-π) > 5**; then X ≈ Normal with mean nπ, variance nπ(1-π).

THE CLASSIC EXAM TRAP "n>30 ⇒ an individual observation is normal" is **FALSE**. The CLT normalizes the **sample mean/sum** only. A single item keeps the population's shape no matter how big n is.

EX · CARS SOLD/DAY: E=1.7, VAR=.81 (SUM)
300 days: sum mean=300(1.7)=510, SD=√(300·.81)=15.59.
P(fewer than 495)= z=(495-510)/15.59=-.96 → **≈ .17**

EX · BRANCH EMPLOYEES M=50, Σ=15 (AVERAGE)
n=40: SE=15/√40=2.37 → z=(55-50)/2.37=2.11 → P($\bar{X}$ >55)=**.017**
n=100: SE=15/√100=1.5 → z=3.33 → P($\bar{X}$ >55)=**.0004**
Bigger n → smaller SE → tighter around μ. Assumes iid, n large.

§8 Descriptive Statistics & Data (Class 01)

Data & sampling vocabulary

- Quantitative** (numbers/arithmetic) vs **Qualitative** (categories). **Discrete** (count) vs **Continuous** (measure).
- Nominal** (no order) vs **Ordinal** (ordered, e.g. Likert). **Cross-sectional** vs **Time series**. **Univariate** vs **Multivariate**.
- Random sample**: every element equally likely to be chosen.

KEY IDEA **Parameter** = summary of the **population** (FIXED). **Statistic** = summary of the **sample** (RANDOM). Populations are fixed; statistics vary sample-to-sample.

Center & spread

SAMPLE / POP MEAN
 $\bar{x} = \sum x/n$ · $\mu = \sum x/N$

MEDIAN
 middle; even $n \rightarrow$ avg of 2

POPULATION VARIANCE
 $\sigma^2 = \sum (x - \mu)^2 / N = (\sum x^2 - N\mu^2) / N$

SAMPLE VARIANCE — DIVIDE BY $n-1$
 $s^2 = \sum (x - \bar{x})^2 / (n-1) = (\sum x^2 - n\bar{x}^2) / (n-1)$

WHY $n-1$? So s^2 is an **unbiased estimator** of σ^2 . Population σ^2 uses N and μ ; sample s^2 uses $n-1$ and $\bar{x}$. Mixing these up is the classic error.

z-scores, quartiles & outliers

$$z = (x - \mu) / \sigma \text{ or } (x - \bar{x}) / s$$

- Outlier by z**: $|z| \geq 3$ outlier; $2 \leq |z| < 3$ possible outlier.
- Empirical rule** (bell-shaped): 68% within $\mu \pm \sigma$, 95% within $\mu \pm 2\sigma$, 99% within $\mu \pm 3\sigma$.
- Quartiles**: Q1 at position $(n+1)/4$; Q3 at $3(n+1)/4$; **IQR=Q3-Q1**.
- Box-plot fences**: inner = $Q1 - 1.5 \cdot \text{IQR}$ & $Q3 + 1.5 \cdot \text{IQR}$ ($\rightarrow$ outlier); outer = $\pm 3 \cdot \text{IQR}$ ($\rightarrow$ serious outlier).

ROBUSTNESS **Sensitive to outliers**: mean, range, s/σ . **Resistant**: median, IQR. Bar chart = qualitative (gaps); histogram = quantitative (no gaps).

EX · BOX PLOT {30,40,70,81,90,118,123}

$Q1=70$, median=81, $Q3=90 \rightarrow \text{IQR}=20$. Fences: $70 - 1.5(20)=40$ & $90 + 1.5(20)=120$. Points outside $\rightarrow$ **outliers: 30 & 123**; whiskers reach 40 & 118.

EX · MEAN VS. MEDIAN (N=9 COMPENSATIONS)

86.24, 88.77, 88.95, 94.62, **95.42**, 95.60, 95.64, 97.18, 99.80 $\rightarrow$ median = middle = **95.42**; mean = 93.58 (pulled down by the low value — mean < median here).

§9 The Sample Midterm — Fully Worked

The game: Roll a fair die. **5 or 6** $\rightarrow$ Jar 1 (15 red, 5 green). **1-4** $\rightarrow$ Jar 2 (8 red, 12 green). So $P(5,6)=2/6=1/3$, $P(1-4)=4/6=2/3$; $P(G|\text{Jar1})=5/20=.25$, $P(G|\text{Jar2})=12/20=.6$.

P1 · P(GREEN BALL) — TOTAL PROBABILITY

$$P(G) = P(G|\text{Jar1})P(5,6) + P(G|\text{Jar2})P(1-4) = .25(1/3) + .6(2/3) = .0833 + .40 = \mathbf{.4833}$$

P2 · P(ROLLED 5-6 | GREEN) — BAYES

$$= P(G|5,6)P(5,6) / P(G) = .0833 / .4833 = \mathbf{.1724}$$

P3 · P(ROLLED 1-4 U GREEN)

$$= P(1-4) + P(G) - P(1-4 \cap G) = .6667 + .4833 - (.6 \cdot .6667) = .6667 + .4833 - .40 = \mathbf{.75}$$

P4 · DISTRIBUTION OF TOTAL PROFIT $Z=X+Y$

Die X: \$2 if 5-6, \$5 if 1-4. Ball Y: \$4 red, \$3 green.

Outcome	Z	P(Z)
5,6 & Green	5	$1/3 \cdot .25 = .0833$
5,6 & Red	6	$1/3 \cdot .75 = .25$
1-4 & Green	8	$2/3 \cdot .6 = .40$
1-4 & Red	9	$2/3 \cdot .4 = .2667$

P5 · E(Z) AND Σ_z

$$E(Z) = 5(.0833) + 6(.25) + 8(.40) + 9(.2667) = \mathbf{7.517}$$

$$E(Z^2) = 25(.0833) + 36(.25) + 64(.40) + 81(.2667) = 58.283$$

$$\text{Var} = 58.283 - 7.517^2 = 1.783 \rightarrow \sigma_z = \mathbf{1.335}$$

P6 · E(XY)

$$= \Sigma(X \cdot Y)P = 6(.0833) + 8(.25) + 15(.40) + 20(.2667) = .5 + 2 + 6 + 5.333 = \mathbf{13.833}$$

P7 · WIN \$9 $\geq 25 \times$ IN 100 PLAYS — BINOMIAL $\approx$ NORMAL

"Win \$9" = roll 1-4 & red, $p=.2667$. $n=100$: mean= $np=26.67$, $\sigma=\sqrt{np(1-p)}=4.42$.

Check $n\pi, n(1-\pi) \geq 5$ ✓. $z=(24.5-26.67)/4.42=-.49 \rightarrow P(X \geq 25) = P(Z > -.49) \approx \mathbf{.69}$ ($1/2$ continuity correction)

P8 · VAR(2X) VS VAR(X+Y) · VAR=49 EACH, INDEP.

$$\text{Var}(2X) = 2^2 \cdot 49 = \mathbf{196}$$
. $\text{Var}(X+Y) = 49 + 49 = \mathbf{98}$ ($\text{Cov}=0$). $\rightarrow$ **Var(2X) > Var(X+Y)**

P9 · INDEPENDENT? $P(A)=.5$, $P(B)=.3$, $P((A \cup B)^c)=.42$

If independent: $P(A \cap B) = .15 \rightarrow P(A \cup B) = .65 \rightarrow P((A \cup B)^c) = .35$. Given $.42 \neq .35 \rightarrow$ **NOT independent**.

P10 · CORN CHIPS: $M=11.95$, $\Sigma=22$, $N=100$

(a) **P(total > 1194)**: sum mean=100(11.95)=1195, $SD=\sqrt{100(.22)}=2.2$. $z=(1194-1195)/2.2=-.4545 \rightarrow P(Z > -.45) = \mathbf{.675}$

(b) **P($\bar{x} < 11.97$)**: $SE=.22/\sqrt{100}=.022$. $z=(11.97-11.95)/.022=.909 \rightarrow \mathbf{.818}$

(c) " $n>30 \Rightarrow$ individual bag is normal"? **No**. CLT normalizes the sample **mean/sum**, not an individual bag — it keeps the population shape for any n .

More Practice · HW2 & Notes Problem Types

OVERBOOKING · EXPECTED VALUE OF A BINOMIAL FUNCTION

40 rooms, 50 reservations, $P(\text{show})=.8 \rightarrow X \sim \text{Bin}(50, .8)$. Pay \$150 per guest bumped ($x > 40$).

$$E[\text{comp}] = 150 \cdot \sum_{x=41..50} (x-40) \cdot P(X=x) = 150(1.119) = \mathbf{\$167.78}$$

POISSON · "AT LEAST 3"

$$\lambda=2 \text{ in the window: } P(X \geq 3) = 1 - [P(0) + P(1) + P(2)] = 1 - e^{-2}(1+2+2) = 1 - .677 = \mathbf{.323}$$

ACCEPTANCE SAMPLING · HYPERGEOMETRIC

Ship $N=200$, inspect 12, reject if **any** defective. With 10 defective:

$$P(\text{reject}) = 1 - C(190, 12) / C(200, 12) = \mathbf{.469}$$
 (binomial approx $1 - (1-10/200)^{12} \approx .460$).

TWO-STAGE BAYES · COIN PICKS A JAR

Jar A 7R/3G, Jar B 15R/30G; fair coin ($H \rightarrow A$, $T \rightarrow B$). $P(\text{red}) = .5(.7) + .5(1/3) = \mathbf{.517}$.

$$P(\text{tails}|\text{green}) = [.5(2/3)] / [.5(.3) + .5(2/3)] = \mathbf{.690}$$

STANDARDIZE TO COMPARE ACROSS CLASSES

Physics 76 ($\mu=70, \sigma=5$) $\rightarrow z=1.2$; History 81 ($80, 3$) $\rightarrow .33$; Stats 78 ($75, 4$) $\rightarrow .75$. Highest $z \Rightarrow$

Physics is best relative to its class.

§10 One-Glance Reference: Formulas, z-Table & Traps

Master formula sheet

Quantity	Formula
Union	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Conditional	$P(A B) = P(A \cap B) / P(B)$
Independent	$P(A \cap B) = P(A)P(B)$
Bayes	$P(B A) = P(A B)P(B) / P(A)$
Combinations	$C(n, k) = n! / [k!(n-k)!]$
$E(X)$	$\sum x \cdot p(x)$
$\text{Var}(X)$	$\sum x^2 p(x) - \mu^2$
$\text{Var}(aX+b)$	$a^2 \cdot \text{Var}(X)$
$\text{Cov}(X, Y)$	$E(XY) - \mu_X \mu_Y = \rho \sigma_X \sigma_Y$
$\text{Var}(aX+bY)$	$a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \cdot \text{Cov}$
Binomial	$C(n, k) \pi^k (1-\pi)^{n-k}$; $\mu = n\pi$, $\sigma^2 = n\pi(1-\pi)$
Poisson	$e^{-\lambda} \lambda^k / k!$; $\mu = \sigma^2 = \lambda$
Hypergeom.	$C(r, x) C(N-r, n-x) / C(N, n)$
Standardize	$z = (x - \mu) / \sigma$
Sum of n	mean $n\mu$, SD $\sigma\sqrt{n}$
Average of n	mean μ , SE $\sigma/\sqrt{n}$
Sample var	$s^2 = \sum (x - \bar{x})^2 / (n-1)$

Standard normal — $A(z) = P(0 \leq Z \leq z)$

z	A(z)	z	A(z)	z	A(z)
0.25	.0987	1.00	.3413	1.75	.4599
0.50	.1915	1.15	.3749	1.96	.4750
0.67	.2486	1.28	.3997	2.00	.4772
0.75	.2734	1.33	.4082	2.33	.4901
0.84	.2995	1.50	.4332	2.50	.4938
0.90	.3159	1.645	.4500	3.00	.4987

Key criticals: $z=1.28 \rightarrow .40$ (90th pct) · $1.645 \rightarrow .45$ (95th) · $1.96 \rightarrow .475$ (97.5th) · $2.33 \rightarrow .49$ (99th). $P(Z \leq z) = .5 + A(z)$.

Excel functions

- =BINOM.DIST(k,n,p,FALSE/TRUE)** — binomial $P(X=k) / P(X \leq k)$
- =POISSON.DIST(k,λ,FALSE/TRUE)**
- =NORM.S.DIST(z,TRUE)** — $\Phi(z) = P(Z \leq z)$ (full cumulative)
- =NORM.DIST(x,μ,σ,TRUE)** · **=COMBIN(n,k)** · **=FACT(n)**

Top exam traps & final checklist

- Variances add, SDs don't.** Never add standard deviations.
- $\text{Var}(aX) = a^2 \text{Var}(X)$** — square the constant; +b does nothing to variance.
- Independent $\Rightarrow \text{Cov}=0$;** if a ρ or Cov is given, you **MUST** include 2·Cov.
- With replacement $\rightarrow$ Binomial; without $\rightarrow$ Hypergeometric.**
- Poisson λ = rate $\times$ the exact time window asked.**
- CLT:** sum uses $\sigma\sqrt{n}$, average uses $\sigma/\sqrt{n}$ — and it never makes an *individual* value normal.
- Sample s^2 divides by $n-1$;** population σ^2 divides by N .
- This class' z-table is "0 $\rightarrow$ z" —** add .5 for a full left tail; use symmetry for negatives.
- Always draw the bell & shade** before computing a normal probability.
- "At least one" = $1 - P(\text{none})$.**